

Sequences**Definition 3.1**

A sequence can be thought of as a list of numbers written in definite order:

$$a_1, a_2, a_3, \dots, a_n, \dots$$

The number a_1 is called the first term, a_2 is the second term, and in general, a_n is the n th term.

Example 3.1

Compute the first 5 terms of the following sequences:

$$(i) \quad a_n = \frac{n}{n+1}, \quad n \geq 1$$

$$a_1 = \frac{1}{2}, a_2 = \frac{2}{3}, a_3 = \frac{3}{4}, a_4 = \frac{4}{5}, a_5 = \frac{5}{6}$$

$$(ii) \quad b_n = \frac{n-1}{n}, \quad n \geq 2$$

$$b_2 = \frac{1}{2}, b_3 = \frac{2}{3}, b_4 = \frac{3}{4}, b_5 = \frac{4}{5}, b_6 = \frac{5}{6}$$

$$(iii) \quad c_n = (-1)^n, \quad n \geq 1$$

$$c_1 = -1, c_2 = 1, c_3 = -1, c_4 = 1, c_5 = -1$$

Definition 3.2

A **geometric progression** is a sequence of the form

$$a, ar, ar^2, \dots, ar^n$$

where the initial term a and the common ratio r are real numbers.

Example 3.2

The sequences

$$b_n = (-1)^n, \quad n \geq 1$$

$$c_n = 2(5)^n, \quad n \geq 1$$

$$d_n = 6\left(\frac{1}{3}\right)^n, \quad n \geq 1$$

are geometric progressions.

UCCM1333 INTRODUCTORY DISCRETE MATHEMATICS

Definition 3.3

An explicit formula (general formula) for a sequence is a rule that shows how the value of a_k depends on k .

Example 3.3

Find an explicit formula for sequences with the following first 5 terms:

(i) $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$

(ii) $1, -1, 1, -1, 1$

(iii) $1, -\frac{1}{4}, \frac{1}{9}, -\frac{1}{16}, \frac{1}{25}$

(iv) $1, 7, 25, 79, 241$

Summation Notation

We use the notation $\sum_{j=m}^n a_j$ to represent $a_m + a_{m+1} + \dots + a_n$.

The variable j is called the index of summation.

The index of summation runs through all integers starting with its lower limit m and ending with its upper limit n .

Note

If we add the terms of a sequence $a_n, n \geq 1$, we get an expression of the form $a_1 + a_2 + a_3 + \dots + a_m = \sum_{n=1}^m a_n$ which is called a **series**.

Example 3.4

(i) $\sum_{i=0}^n \frac{(-1)^i}{i+1}$

(ii) $\frac{1}{n} + \frac{2}{n+1} + \frac{3}{n+2} + \dots + \frac{n+1}{2n}$

(ii) Compute $\sum_{k=1}^n \left(\frac{k}{k+1} - \frac{k+1}{k+2} \right)$

(iii) What is the value of $\sum_{k=2}^5 k^2$?

Theorem 3.1

If a and r are real numbers and $r \neq 0$, then

$$\sum_{j=0}^n ar^j = \begin{cases} \frac{ar^{n+1} - a}{r - 1}, & r \neq 1 \\ (n+1)a, & r = 1 \end{cases}$$

Change of Variable

The symbol used to represent the index of a summation can be replaced

by any other symbol. For example, $\sum_{j=1}^5 j^2 = \sum_{k=0}^4 (k+1)^2$.

Example 3.5

Transform the following summation by making the specified change of variable:

(i) $\sum_{j=0}^6 \frac{1}{j+1}$, change of variable: $k = j + 1$

(ii) $\sum_{j=1}^{n+1} n - j + 1$, change of variable: $k = j - 1$.

(iii) $\sum_{j=1}^{n+1} \frac{j}{n+j}$, change of variable: $k = j - 1$.

Example 3.6

Double Summations:

$$\sum_{i=1}^4 \sum_{j=1}^3 ij =$$

Product Notation

Π denotes a product.

$$\prod_{k=m}^n a_k = a_m \cdot a_{m+1} \cdot a_{m+2} \cdots a_n$$

Example 3.7

Compute the following products:

(i) $\prod_{k=1}^1 \frac{k}{k+1} =$

(ii) $\prod_{k=1}^3 k(k+1) =$

(iii) $\prod_{k=1}^5 \frac{1}{k} =$

Note

(i) $\sum_{n=1}^m a_n + \sum_{n=1}^m b_n = \sum_{n=1}^m (a_n + b_n)$

(ii) $\sum_{n=1}^m ca_n = c \sum_{n=1}^m a_n$

(iii) $\left(\prod_{n=1}^m a_n \right) \left(\prod_{n=1}^m b_n \right) = \prod_{n=1}^m (a_n b_n)$

UCCM1333 INTRODUCTORY DISCRETE MATHEMATICS

Factorial Notation

For each positive integer n , the quantity n factorial, denoted $n!$, is defined to be the product of all the integers from 1 to n .

$$n! = n(n-1)(n-2)\dots(3)(2)(1)$$

Zero factorial is defined to be 1.

The following formula holds for each positive integer n :

$$n! = n(n-1)!$$

Example 3.8

Simplify the following expressions:

$$(i) \quad \frac{1}{2!4!} + \frac{1}{3!3!} =$$

$$(ii) \quad \frac{(n+1)!}{n!} =$$

$$(ii) \quad \frac{n!}{(n-3)!} =$$

Sequences In Computer Programming

An important data type in computer programming consists of finite sequences. In computer programming contexts, these are usually referred to as *one-dimensional arrays*. For example, consider a program that analyzes the wages paid to a sample of 50 workers. Such a program might compute the average wage and the difference between each individual wage and the average. This would require that each wage be stored in memory for retrieval later in the calculation. To avoid the use of entirely separate variable names for all of the 50 wages, each is written as a term of a one-dimensional array:

$$W[1], W[2], W[3], \dots, W[50].$$

Note that the subscript labels are written inside square brackets. The reason is that until relatively recently, it was impossible to type a actual dropped subscripts on most computer keyboards.

UCCM1333 INTRODUCTORY DISCRETE MATHEMATICS

Example 3.9 (Dummy Variable in a Loop)

The index variable for a **for-next** loop is a **dummy variable**. For example, the following three algorithm segments all produce the same output:

- 1) **for** $i := 1$ **to** n
 print $a[i]$
 next i
- 2) **for** $j := 0$ **to** $n - 1$
 print $a[j + 1]$
 next j
- 3) **for** $k := 2$ **to** $n + 1$
 print $a[k - 1]$
 next k

The recursive definitions for summation, product, and factorial lead naturally to computational algorithms. For instance, here are two sets of pseudocode to find the sum of $a[1], a[2], \dots, a[n]$.

$s := a[1]$	$s := 0$
for $k := 2$ to n	for $k := 1$ to n
$s := s + a[k]$	$s := s + a[k]$
next k	next k

Mathematical Induction

Mathematical induction is one of the more recently developed techniques of proof in the history of mathematics. It is used to check conjectures about the outcomes of processes that occur repeatedly and according to definite patterns.

Principle of Ordinary Mathematical Induction

Let $P(n)$ be a predicate that is defined for integers n , and let a be a fixed integer. Suppose the following two statements are true:

- (1) $P(a)$ is true.
- (2) For all integers $k \geq a$, if $P(k)$ is true then $P(k+1)$ is true.

Then the statement “for all integers $n \geq a$, $P(n)$ ” is true.

Proving a statement by mathematical induction is a two-step process:

Method of Proof by Mathematical Induction

UCCM1333 INTRODUCTORY DISCRETE MATHEMATICS

Consider a statement of the form, “For all integers $n \geq a$, a property $P(n)$ is true.”

To prove such a statement, perform the following two steps:

Step 1: Basis Step

Show that the property is true for $n = a$.

Step 2: Inductive Step

Show that for all integers $k \geq a$, if the property is true for $n = k$ then it is true for $n = k + 1$.

To perform this step,

Suppose that the property is true for $n = k$ for some integer $k \geq a$.
[this supposition is called the **inductive hypothesis**]

Then

Show that the property is true for $n = k + 1$.

Note

In a proof by mathematical induction, it is not assumed that $P(k)$ is true for all positive integers!

Examples Of Proofs By Mathematical Induction

Example 3.10

Use mathematical induction to prove that the sum of the first n odd positive integers is n^2 .

Let $P(n)$: The sum of the first n odd positive integers is n^2 .

or $P(n)$: $1 + 3 + 5 + \dots + (2n - 1) = n^2$, $n \geq 1$.

Basis step:

$n = 1$, LHS = 1, RHS = 1

LHS = RHS

So $P(1)$.

Inductive step:

Suppose that $P(k)$ is true for a positive integer $k \geq 1$; that is,

$$1 + 3 + 5 + \dots + (2k - 1) = k^2.$$

Show $P(k + 1)$ is true: $1 + 3 + \dots + (2(k + 1) - 1) = (k + 1)^2$

When $n = k + 1$,

UCCM1333 INTRODUCTORY DISCRETE MATHEMATICS

$$\begin{aligned}
1 + 3 + 5 + \dots + (2k - 1) + (2(k + 1) - 1) &= k^2 + (2(k + 1) - 1) \\
&= k^2 + 2k + 1 \\
&= (k + 1)^2.
\end{aligned}$$

Hence $P(k + 1)$ is true.

By mathematical induction, $P(n)$ is true for all $n \geq 1$.

(Sum of the first n integers)**Example 3.11**

Use mathematical induction to prove that $\sum_{r=1}^n r = 1 + 2 + \dots + n = \frac{n(n+1)}{2}$

Let $P(n)$: $\sum_{r=1}^n r = \frac{n(n+1)}{2}$

Basis Step:

When $n = 1$, LHS = 1, RHS = $\frac{1(1+1)}{2} = 1$

So LHS = RHS and $P(1)$ is true.

Inductive Step:

Suppose $P(k)$ is true for a positive integer $k \geq 1$ that is

$$1 + 2 + \dots + k = \frac{k(k+1)}{2}.$$

Show $P(k + 1)$ is true: $1 + 2 + \dots + (k + 1) = \frac{(k+1)(k+2)}{2}$

When $n = k + 1$,

$$\begin{aligned}
1 + 2 + \dots + k + (k + 1) &= \frac{k(k+1)}{2} + (k + 1) \\
&= \frac{(k+1)(k+2)}{2}.
\end{aligned}$$

So $P(k + 1)$ is true.

By mathematical induction, $P(n)$ is true for all $n \geq 1$.

Example 3.12

Use mathematical induction to prove that

$$\sum_{r=1}^n r^2 = 1^2 + 2^2 + \dots + n^2 = \frac{1}{6}n(n+1)(2n+1).$$

Let $P(n)$: $1^2 + 2^2 + \dots + n^2 = \frac{1}{6}n(n+1)(2n+1)$

Example 3.13

Use mathematical induction to prove that

$$\sum_{r=1}^n r^3 = 1^3 + 2^3 + \dots + n^3 = \left[\frac{1}{2}n(n+1) \right]^2.$$

Let $P(n)$: $\sum_{r=1}^n r^3 = 1^3 + 2^3 + \dots + n^3 = \left[\frac{1}{2}n(n+1) \right]^2$

Applying the Formula for the Sum of the First n Integers

$$1) \quad \sum_{r=1}^n r = \frac{n(n+1)}{2}$$

$$2) \quad \sum_{r=1}^n r^2 = \frac{1}{6}n(n+1)(2n+1)$$

$$3) \quad \sum_{r=1}^n r^3 = \left[\frac{1}{2}n(n+1) \right]^2$$

Example 3.14

Write down the k^{th} term of the given series and deduce its sum to n terms.

(a) $1(1) + 2(3) + 3(5) + \dots$

k^{th} term of the series, $U_k = k(2k-1) = 2k^2 - k$.

$$\begin{aligned} \sum_{r=1}^n U_r &= 2 \sum_{r=1}^n r^2 - \sum_{r=1}^n r \\ &= 2 \left[\frac{1}{6}n(n+1)(2n+1) \right] - \frac{1}{2}n(n+1) \\ &= \frac{1}{6}n(n+1)[4n+2-3] \\ &= \frac{1}{6}n(n+1)(4n-1) \end{aligned}$$

(b) $1(3) + 3(5) + 5(7) + \dots$

(c) $1(2) + 2(3) + 3(4) + \dots$

(d) $2(5) + 5(8) + 8(11) + \dots$

(e) $1(4) + 2(7) + 3(10) + \dots$

Proving a Divisibility Property

Example 3.16

Use mathematical induction to prove that for all integers $n \geq 1$, $2^{2n} - 1$ is divisible by 3.

Let $P(n)$: $2^{2n} - 1$ is divisible by 3, $n \geq 1$

Basis Step:

$n = 1$, $2^{2(1)} - 1 = 3$ is divisible by 3.

So $P(1)$ is true.

Inductive Step:

Suppose $P(k)$ is true for a positive integer $k \geq 1$ that is

$$2^{2k} - 1 \text{ is divisible by 3}$$

We must show that $2^{2(k+1)} - 1$ is divisible by 3.

When $n = k + 1$,

$$\begin{aligned} 2^{2(k+1)} - 1 &= 2^{2k+2} - 1 \\ &= 4(2^{2k}) - 1 \\ &= 3(2^{2k}) + (2^{2k} - 1). \end{aligned}$$

By the inductive hypothesis, $2^{2k} - 1$ is divisible by 3 and clearly $(3)2^{2k}$ is divisible by 3. Thus $P(k + 1)$ is true.

Hence, by mathematical induction, $P(n)$ is true for all integers $n \geq 1$.

UCCM1333 INTRODUCTORY DISCRETE MATHEMATICS

Example 3.17

Use mathematical induction to prove that for all integers $n \geq 1$, $7^n(3n+1) - 1$ is divisible by 9.

Let $P(n)$: $7^n(3n+1) - 1$ is divisible by 9, $n \geq 1$.

Proving an Inequality

Example 3.18

Use mathematical induction to prove that $2n + 1 < 2^n$, for all integers $n \geq 3$.

Let $P(n)$: $2n + 1 < 2^n$, $n \geq 3$.

Basis Step:

$n = 3$, LHS = $2(3) + 1 = 7$, RHS = $2^3 = 8$.

Hence LHS < RHS and so $P(3)$ is true.

Inductive Step:

Suppose $P(k)$ is true for some integer $k \geq 3$, that is

$$2k + 1 < 2^k$$

Show that $P(k + 1)$ is true: $2(k+1) + 1 < 2^{k+1}$.

When $n = k + 1$,

$$2(k+1) = 2k + 2$$

$$= (2k + 1) + 2$$

$$< 2^k + 2$$

$$< 2^k + 2^k = 2(2^k) = 2^{k+1} \text{ (since } 2 < 2^k \text{ for } k \geq 3)$$

So $P(k + 1)$ is true.

Hence, by mathematical induction, $P(n)$ is true for all integers $n \geq 3$.

UCCM1333 INTRODUCTORY DISCRETE MATHEMATICS

Example 3.19

Use mathematical induction to prove the inequality $n < 2^n$ for all positive integers n .

Let $P(n)$: $n < 2^n$, $n \geq 1$.

Proving a property of a sequence

Example 3.20

Define a sequence $a_1, a_2, \dots$ as follows:

$$a_1 = 2 \text{ and } a_k = 5a_{k-1} \text{ for all integers } k \geq 2.$$

Use mathematical induction to show that the terms of the sequence satisfy the formula

$$a_n = 2(5^{n-1}) \text{ for all integers } n \geq 1.$$

Let $P(n)$: $a_n = 2(5^{n-1})$, $n \geq 1$

Basis Step:

$$n = 1, a_1 = 2(5^{1-1}) = 2$$

By the definition of the sequence, $a_1 = 2$, so $P(1)$ is true.

Inductive Step:

Suppose $P(k)$ is true for some integer $k \geq 1$, that is

$$a_k = 2(5^{k-1})$$

Show that $P(k + 1)$ is true: $a_{k+1} = 2(5^{(k+1)-1})$

When $n = k + 1$,

UCCM1333 INTRODUCTORY DISCRETE MATHEMATICS

$$\begin{aligned}a_{k+1} &= 5a_{(k+1)-1} \\&= 5a_k \\&= 5[2(5^{k-1})] \\&= 2(5^k)\end{aligned}$$

So $P(k+1)$ is true.

Hence, by mathematical induction, $P(n)$ is true for all integers $n \geq 1$.

Sum of a Geometric Sequence

Use mathematical induction to prove that $\sum_{i=0}^n r^i = \frac{r^{n+1}-1}{r-1}$ for all integers $n \geq 0$ and all real numbers r except 1.

$$\text{Let } P(n): \sum_{i=0}^n r^i = \frac{r^{n+1}-1}{r-1}$$

Basis Step:

$$n = 0, \text{ LHS} = r^0 = 1, \quad \text{RHS} = \frac{r^{0+1}-1}{r-1} = 1$$

So LHS = RHS, $P(1)$ is true.

Inductive Step:

Suppose $P(k)$ is true for some integer $k \geq 1$, that is

$$\sum_{i=0}^k r^i = \frac{r^{k+1}-1}{r-1}$$

$$\text{Show that } P(k+1) \text{ is true: } \sum_{i=0}^{k+1} r^i = \frac{r^{k+2}-1}{r-1}$$

When $n = k+1$,

$$\begin{aligned}\sum_{i=0}^{k+1} r^i &= \sum_{i=0}^k r^i + r^{(k+1)} \\&= \frac{r^{k+1}-1}{r-1} + r^{k+1} \\&= \frac{r^{k+1}-1 + r^{k+1}(r-1)}{r-1} \\&= \frac{r^{k+1}-1 + r^{k+2} - r^{k+1}}{r-1} \\&= \frac{r^{k+2}-1}{r-1}\end{aligned}$$

So $P(k+1)$ is true. Hence, by mathematical induction, $P(n)$ is true for all integers $n \geq 0$.

Example 3.21

Let m be an integer greater than or equal to 3.

- (i) Find $1 + 3 + 3^2 + \dots + 3^{m-2}$.
 (ii) Find $3^2 + 3^3 + 3^4 + \dots + 3^m$.

$$(i) \quad 1 + 3 + 3^2 + \dots + 3^{m-2} = \sum_{i=0}^{m-2} 3^i = \frac{3^{m-2+1} - 1}{3 - 1} = \frac{3^{m-1} - 1}{2}$$

$$(ii) \quad 3^2 + 3^3 + 3^4 + \dots + 3^m = 3^2(1 + 3 + 3^2 + \dots + 3^{m-2})$$

$$= 3^2 \left(\frac{3^{m-1} - 1}{2} \right) = \frac{3^{m+1} - 9}{2}$$

Method of Differences

If $U_r = f(r) - f(r-1)$, then

$$\sum_{r=1}^n U_r = \sum_{r=1}^n \{f(r) - f(r-1)\}$$

$$= + \begin{bmatrix} f(1) & - & f(0) \\ f(2) & - & f(1) \\ \vdots & & \\ f(n) & - & f(n-1) \end{bmatrix}$$

$$= f(n) - f(0)$$

Example 3.22

Find $1(2)(3) + 2(3)(4) + \dots + n(n+1)(n+2)$.

Let $U_r = r(r+1)(r+2)$.

Let $f(r) = r(r+1)(r+2)(r+3)$ and $f(r-1) = (r-1)r(r+1)(r+2)$.

$$f(r) - f(r-1) = r(r+1)(r+2)(r+3) - (r-1)r(r+1)(r+2)$$

$$= r(r+1)(r+2)[(r+3) - (r-1)]$$

$$= 4U_r$$

$$\text{So } U_r = \frac{1}{4}[f(r) - f(r-1)]$$

$$\begin{aligned}
\sum_{r=1}^n U_r &= \frac{1}{4} \sum_{r=1}^n \{f(r) - f(r-1)\} \\
&= \frac{1}{4} \left\{ + \begin{bmatrix} f(1) & - & f(0) \\ f(2) & - & f(1) \\ \vdots & & \\ f(n) & - & f(n-1) \end{bmatrix} \right\} \\
&= \frac{1}{4} [f(n) - f(0)] \\
&= \frac{1}{4} n(n+1)(n+2)(n+3).
\end{aligned}$$

Example 3.23

Sum to n terms $2(5) + 5(8) + 8(11) + \dots$

Let $U_r = (3r - 1)(3r + 2)$.

Let $f(r) = (3r - 1)(3r + 2)(3r + 5)$ and $f(r - 1) = (3r - 4)(3r - 1)(3r + 2)$.

$$\begin{aligned}
\text{Then } f(r) - f(r - 1) &= (3r - 1)(3r + 2)(3r + 5) - (3r - 4)(3r - 1)(3r + 2) \\
&= (3r - 1)(3r + 2)(3r + 5 - (3r - 4)) \\
&= 9U_r.
\end{aligned}$$

Hence

$$\begin{aligned}
\sum_{r=1}^n U_r &= \frac{1}{9} \sum_{r=1}^n \{f(r) - f(r-1)\} \\
&= \frac{1}{9} \left\{ + \begin{bmatrix} f(1) & - & f(0) \\ f(2) & - & f(1) \\ \vdots & & \\ f(n) & - & f(n-1) \end{bmatrix} \right\} \\
&= \frac{1}{9} [f(n) - f(0)] \\
&= \frac{1}{9} [(3n - 1)(3n + 2)(3n + 5) + 10].
\end{aligned}$$

Example 3.24

Let $f(r) = \frac{1}{4}r^2(r+1)^2$, show that $f(r) - f(r-1) = r^3$.

Hence find the sum of the series $1^3 + 2^3 + 3^3 + \dots + n^3$.

$$\begin{aligned} f(r) &= \frac{1}{4}r^2(r+1)^2, & f(r-1) &= \frac{1}{4}(r-1)^2(r)^2 \\ f(r) - f(r-1) &= \frac{1}{4}r^2((r+1)^2 - (r-1)^2) \\ &= \frac{1}{4}r^2(r^2 + 2r + 1 - r^2 + 2r - 1) \\ &= r^3 \end{aligned}$$

$$\begin{aligned} 1^3 + 2^3 + 3^3 + \dots + n^3 &= \sum_{r=1}^n r^3 \\ &= \sum_{r=1}^n [f(r) - f(r-1)] \\ &= + \begin{bmatrix} f(1) & - & f(0) \\ f(2) & - & f(1) \\ & \vdots & \\ f(n) & - & f(n-1) \end{bmatrix} \\ &= f(n) - f(0) \\ &= \frac{1}{4}n^2(n+1)^2 \end{aligned}$$

Example 3.25

Let $f(r) = r(r+1)!$. Simplify $f(r) - f(r-1)$ and hence sum to n terms the series $5(2!) + 10(3!) + 17(4!) + 26(5!) + \dots$

$$\begin{aligned} f(r) &= r(r+1)! & f(r-1) &= (r-1)(r-1+1)! = (r-1)r! \\ f(r) - f(r-1) &= r(r+1)! - (r-1)r! \\ &= r(r+1)r! - (r-1)r! \\ &= r![r^2 + r - (r-1)] \\ &= (r^2 + 1)r! \end{aligned}$$

$$\begin{aligned} 5(2!) + 10(3!) + 17(4!) + 26(5!) + \dots + (n^2 + 1)n! \\ &= \sum_{r=2}^n (r^2 + 1)r! \\ &= \sum_{r=2}^n [f(r) - f(r-1)] \end{aligned}$$

$$\begin{aligned}
 &= + \begin{bmatrix} f(2) & - & f(1) \\ f(3) & - & f(2) \\ \vdots & & \\ f(n) & - & f(n-1) \end{bmatrix} \\
 &= f(n) - f(1) \\
 &= n(n+1)! - 2
 \end{aligned}$$

Fractions

Example 3.26

Sum to n terms the series $\frac{1}{3(5)} + \frac{1}{5(7)} + \frac{1}{7(9)} + \dots$

$$\text{Let } U_r = \frac{1}{(2r+1)(2r+3)}$$

$$\text{Let } f(r) = \frac{1}{2r+3} = (2r+1)U_r$$

Then

$$f(r-1) = \frac{1}{2(r-1)+3} = \frac{1}{2r+1} = (2r+3)U_r$$

$$\begin{aligned}
 f(r) - f(r-1) &= U_r[2r+1 - (2r+3)] \\
 &= -2U_r
 \end{aligned}$$

$$\frac{1}{3(5)} + \frac{1}{5(7)} + \frac{1}{7(9)} + \dots + \frac{1}{(2n+1)(2n+3)}$$

$$= \sum_{r=1}^n \frac{1}{(2r+1)(2r+3)}$$

$$= \sum_{r=1}^n U_r$$

$$= \sum_{r=1}^n -\frac{1}{2}[f(r) - f(r-1)]$$

$$= -\frac{1}{2} \left\{ + \begin{bmatrix} f(1) & - & f(0) \\ f(2) & - & f(1) \\ \vdots & & \\ f(n) & - & f(n-1) \end{bmatrix} \right\}$$

$$= -\frac{1}{2}(f(n) - f(0))$$

$$= -\frac{1}{2} \left[\frac{1}{2n+3} - \frac{1}{3} \right]$$

$$= \frac{n}{3(2n+3)}$$

Example 3.27

Sum to n terms the series $\frac{1}{3(5)(7)} + \frac{1}{5(7)(9)} + \frac{1}{7(9)(11)} + \dots$

$$\text{Let } U_r = \frac{1}{(2r+1)(2r+3)(2r+5)}$$

$$\text{Let } f(r) = \frac{1}{(2r+3)(2r+5)} = (2r+1)U_r, \text{ then}$$

$$f(r-1) = \frac{1}{(2r+1)(2r+3)} = (2r+5)U_r$$

$$f(r) - f(r-1) = (2r+1)U_r - (2r+5)U_r$$

$$= -4U_r$$

$$\frac{1}{3(5)(7)} + \frac{1}{5(7)(9)} + \frac{1}{7(9)(11)} + \dots + \frac{1}{(2n+1)(2n+3)(2n+5)}$$

$$= \sum_{r=1}^n \frac{1}{(2r+1)(2r+3)(2r+5)}$$

$$= -\frac{1}{4} \sum_{r=1}^n [f(r) - f(r-1)]$$

$$= -\frac{1}{4} \left\{ \begin{array}{l} \left[\begin{array}{cc} f(1) & - & f(0) \\ f(2) & - & f(1) \\ \vdots & & \\ f(n) & - & f(n-1) \end{array} \right] \end{array} \right\}$$

$$= -\frac{1}{4} (f(n) - f(0))$$

$$= -\frac{1}{4} \left[\frac{1}{(2n+3)(2n+5)} - \frac{1}{15} \right]$$

$$= -\frac{1}{4(15)} \left[\frac{15 - 4n^2 - 16n - 15}{(2n+3)(2n+5)} \right]$$

$$= -\frac{1}{4(15)} \left[\frac{15 - 4n^2 - 16n - 15}{(2n+3)(2n+5)} \right] = \frac{n(n+4)}{15(2n+3)(2n+5)}$$

Program Verification

Suppose that we have designed an algorithm to solve a problem and have written a program to implement it.

How can we be sure that the program always produces the correct answer?

After all bugs have been removed so that the syntax is correct, we can test the program with sample input.

It is not correct if an incorrect result is produced for any sample input. But even if the program gives the correct answer for all sample input, it may not always produce the correct answer (unless all possible input has been tested).

We need a proof to show that the program *always* gives the correct output.

Definition 3.4

A program is said to be **correct** if it produces the correct output for every possible input.

A proof that a program is correct consists of two parts.

- (1) establishes the partial correctness of the program, (show that the correct answer is obtained if the program terminates)
- (2) shows that the program always terminates.

To specify what it means for a program to produce the correct output, two propositions are used.

- (1) The **initial assertion** gives the properties that the input values must have.
- (2) The **final assertion** gives the properties that the output of the program should have, if the program did what was intended.

The appropriate initial and final assertions must be provided when a program is checked.

Definition 3.5

A program, or program segment S , is said to be **partially correct with respect to** the initial assertion p and the final assertion q if whenever p is true for the input values of S and S terminates, then q is true for the output values of S .

Notation

$p\{S\}q$ indicates that the program, or program segment, S is partially correct with respect to the initial assertion p and the final assertion q .

Example 3.28

Show that the program segment

$$\begin{array}{l} y := 2 \\ z := x + y \end{array}$$

is correct with respect to the initial assertion $p: x = 1$ and the final assertion $q: z = 3$.

Suppose that p is true, so that $x = 1$ as the program begins. Then y is assigned the value 2 and z is assigned the sum of the values of x and y , which is 3. Hence, S is correct with respect to the initial assertion p and the final assertion q . Thus $p\{S\}q$ is true.

Conditional statement

Suppose that a program segment has the form

$$\begin{array}{c} \text{if } \textit{condition} \text{ then} \\ S \end{array}$$

where S is a block of statements.

S is executed if *condition* is true, and it is not executed when *condition* is false

To verify that this segment is correct with respect to the initial assertion p and the final assertion q , show that

- (i) when p is true and *condition* is also true, then q is true after S terminates;
- (ii) when p is true and *condition* is false, then q is true. since in this case S does not execute.

Example 3.29

Verify that the program segment

$$\begin{array}{l} \text{if } x > y \text{ then} \\ y := x \end{array}$$

is correct with respect to the initial assertion T and the final assertion $y \geq x$.

When the initial assertion is true and $x > y$, the assignment $y := x$ is carried out. Hence, the final assertion, which asserts that $y \geq x$, is true in this case. Moreover, when the initial assertion is true and $x > y$ is false, so that $x \leq y$, the final assertion is again true. So the program is correct with respect to the given initial and final assertions.

Similarly, suppose that a program has a statement of the form

if condition then

S_1

else

S_2

If *condition* is true, then S_1 executes; if *condition* is false, then S_2 executes.

To verify that this program segment is correct with respect to the initial assertion p and the final assertion q , show that

- (i) when p is true and *condition* is also true, then q is true after S_1 terminates;
- (ii) when p is true and *condition* is false, then q is true after S_2 terminates.

Example 3.30

Verify that the program segment

if $x < 0$ then

$\text{abs} := -x$

else

$\text{abs} := x$

is correct with respect to the initial assertion T and the final assertion $\text{abs} = |x|$.

Two things must be demonstrated.

- (1) it must be shown that if the initial assertion is true and $x < 0$, then $\text{abs} = |x|$.

This is correct, since when $x < 0$ the assignment statement $\text{abs} := -x$ sets $\text{abs} = -x$, which is $|x|$ by definition when $x < 0$.

- (2) it must be shown that if the initial assertion is true and $x < 0$ is false, so that $x \geq 0$, then $\text{abs} = |x|$.

This is also correct, since in this case the program uses the assignment statement $\text{abs} := x$, and x is $|x|$ by definition when $x \geq 0$, so that $\text{abs} := x$.

Hence this segment is correct with respect to the given initial and final assertions.

Example 3.31

In the program segment shown below, show that if the predicate is true before entry to the loop, then it is also true after exit from the loop.

Loop: **while** ($m \geq 0$ and $m \leq 100$)
 $m := m + 1$
 $n := n - 1$
 end while
predicate: $m + n = 100$

Suppose the condition $m + n = 100$ is true before entry to the loop. Then
 $m_{\text{old}} + n_{\text{old}} = 100$

Show: $m_{\text{new}} + n_{\text{new}} = 100$

After execution of the loop,

$$m_{\text{new}} = m_{\text{old}} + 1$$

$$n_{\text{new}} = n_{\text{old}} - 1$$

$$\text{So } m_{\text{new}} + n_{\text{new}} = (m_{\text{old}} + 1) + (n_{\text{old}} - 1) = m_{\text{old}} + n_{\text{old}} = 100$$

Recursively Define Sequences

Definition 3.6

- (i) A sequence is **defined recursively** if certain initial values are specified and later terms of the sequence are defined by relating them to a number of earlier terms.
- (ii) A **recurrence relation** for a sequence $a_0, a_1, a_2, \dots$ is a formula that relates each term a_k to certain of its predecessors $a_{k-1}, a_{k-2}, \dots, a_{k-i}$, where i is an integer and k is any integer greater than or equal to i .

Example 3.32

Define a sequence $b_0, b_1, b_2, \dots$ recursively as follows:

For all integers $k \geq 2$,

$$b_k = b_{k-1} + kb_{k-2} + 1 \quad \text{(recurrence relation)}$$

$$b_0 = 1, b_1 = 2 \quad \text{(initial conditions)}$$

Find b_2, b_3 and b_4 .

$$\begin{aligned} b_2 &= b_{2-1} + 2b_{2-2} + 1 \\ &= b_1 + 2b_0 + 1 \\ &= 2 + 2(1) + 1 \\ &= 5 \end{aligned}$$

$$\begin{aligned} b_3 &= b_{3-1} + 2b_{3-2} + 1 \\ &= b_2 + 2b_1 + 1 \\ &= 5 + 2(2) + 1 \\ &= 9 \end{aligned}$$

$$\begin{aligned} b_4 &= b_{4-1} + 2b_{4-2} + 1 \\ &= b_3 + 2b_2 + 1 \\ &= 9 + 2(5) + 1 \\ &= 19 \end{aligned}$$

Example 3.33

Define a sequence $c_0, c_1, c_2, \dots$ recursively as follows:

For all integers $k \geq 1$,

$$c_k = 3c_{k-1} \quad \text{and} \quad c_0 = 2.$$

Find c_1, c_2 and c_3 .

$$\begin{aligned} c_1 &= 3c_{1-1} = 3c_0 = 3(2) = 6 \\ c_2 &= 3c_{2-1} = 3c_1 = 3(6) = 18 \\ c_3 &= 3c_{3-1} = 3c_2 = 3(18) = 54 \end{aligned}$$

Showing a sequence given by an explicit formula satisfies certain recurrence relation

Example 3.34

Show that the sequence $1, -1!, 2!, -3!, 4!, -5!, 6!, \dots, (-1)^n n!, \dots$, for $n \geq 0$, satisfies the recurrence relation $s_k = (-k)s_{k-1}$ for all integers $k \geq 1$.

The explicit formula for the sequence is $s_n = (-1)^n n!$

$$\begin{aligned} (-k)s_{k-1} &= (-k)[(-1)^{k-1}(k-1)!] \\ &= (-1)(-1)^{k-1}[k(k-1)!] \\ &= (-1)^k k! \\ &= s_k \end{aligned}$$

The Fibonacci Numbers

Example 3.35

For all integers $k \geq 2$, the Fibonacci sequence is defined recursively as follows:

$$F_k = F_{k-1} + F_{k-2} \quad (\text{recurrence relation})$$

$$F_0 = 1, F_1 = 1. \quad (\text{initial condition})$$

- (i) Find the Fibonacci numbers F_2, F_3, F_4, F_5 and F_6 .
- (ii) Prove that $F_k = 3F_{k-3} + 2F_{k-4}$ for all integers $k \geq 4$.

Compound Interest

Example 3.36

RM100,000 is deposited in a bank account earning 4% interest compounded annually. What is the amount in the account after 5 years?

For each positive integer n ,

Let A_n = the amount in the account at the end of year n

and A_0 = the initial amount in the account = 100 000

The sequence $A_0, A_1, A_2, \dots$ is defined as follows:

For all integers $k \geq 1$,

$$A_k = (1.04) A_{k-1} \quad (\text{recurrence relation})$$

$$A_0 = 100000 \quad (\text{initial condition})$$

General Recursive Definitions.

Recursively defined Sets

A recursive definition for a set consists of the following three components:

- I. **Base:** A statement that certain objects belong to the set.
- II. **Recursion:** A collection of rules indicating how to form new set objects from those already known to be in the set.
- III. **Restriction:** A statement that no objects belong to the set other than those coming from 1 and 2.

Example 3.37

Consider the set S of all strings in a 's and b 's. S is defined recursively as follows:

- I. Base: λ is in S , where λ is the null string.
- II. Recursion: If $s \in S$, then
 - (a) $sa \in S$ and
 - (b) $sb \in S$,
- III. Restriction: Nothing is in S other than objects defined in 1 and 2 above.

Derive the fact that $ab \in S$.

Example 3.38

Below is a recursive definition to generate the set P of legal configuration of parentheses.

- I. Base: $()$ is in P
- II. Recursion:
 - (a) if E is in P , so is (E) .
 - (b) if E and F are in P , so is EF .
- III. Restriction: No configurations of parentheses are in P other than those derived from 1 and 2 above.

Derive the fact that $(())()$ is in P .

Recursive Definition of sum, product, union and intersection.

Definition 3.7

Given numbers $a_1, a_2, a_3, \dots, a_n$ where n is a positive integer, the summation from $i=1$ to n of the a_i , denoted $\sum_{i=1}^n a_i$, is defined by

$$\sum_{i=1}^1 a_i = a_1 \text{ and } \sum_{i=1}^n a_i = \left(\sum_{i=1}^{n-1} a_i \right) + a_n \text{ if } n > 1.$$

The product from $i=1$ to n of the a_i , denoted $\prod_{i=1}^n a_i$, is defined by

$$\prod_{i=1}^1 a_i = a_1 \text{ and } \prod_{i=1}^n a_i = \left(\prod_{i=1}^{n-1} a_i \right) \cdot a_n \text{ if } n > 1.$$

Definition 3.8

Given sets $A_1, A_2, \dots, A_n$, where n is a positive integer, the union of the A_i from $i=1$ to n , denoted $\bigcup_{i=1}^n A_i$ is defined by

$$\bigcup_{i=1}^1 A_i = A_1 \text{ and } \bigcup_{i=1}^n A_i = \left(\bigcup_{i=1}^{n-1} A_i \right) \cup A_n.$$

The intersection of the A_i from $i=1$ to n , denoted $\bigcap_{i=1}^n A_i$, is defined by

$$\bigcap_{i=1}^1 A_i = A_1 \text{ and } \bigcap_{i=1}^n A_i = \left(\bigcap_{i=1}^{n-1} A_i \right) \cap A_n.$$

Section 3.6 Solving Recurrence Relations by Iteration

The most basic method for finding an explicit formula for recursively defined sequence is iteration.

Iteration works as follows:

Given a sequence $a_0, a_1, a_2, \dots$ defined by a recurrence relation and initial conditions, we start from the initial conditions and calculate successive terms of the sequence until we see a pattern developing. At that point we guess an explicit formula.

Example 3.39

Let $a_0, a_1, a_2, \dots$ be the sequence defined recursively as follows:

For all integers $k \geq 1$,

$$a_k = a_{k-1} + 2$$

$$a_0 = 1$$

Use iteration to guess an explicit formula for the sequence.

$$a_0 = 1$$

$$a_1 = a_0 + 2 = 1 + 2$$

$$a_2 = a_1 + 2 = 1 + 2 + 2 = 1 + 2(2)$$

$$a_3 = a_2 + 2 = 1 + 2 + 2 + 2 = 1 + 3(2)$$

$$a_4 = a_3 + 2 = 1 + 2 + 2 + 2 + 2 = 1 + 4(2)$$

$\vdots$

In general, $a_n =$

The sequence obtained in Example 3.39 is called an **arithmetic sequence** which is defined as follows:

Definition 3.9

A sequence $a_0, a_1, a_2, \dots$ is called an arithmetic sequence if and only if there is a constant d such that

$$a_k = a_{k-1} + d \text{ for all integers } k \geq 1.$$

Or equivalently,

$$a_n = a_0 + dn \text{ for all integers } n \geq 0.$$

UCCM1333 INTRODUCTORY DISCRETE MATHEMATICS

Example 3.40

Let r be a fixed nonzero constant, and suppose a sequence $a_0, a_1, a_2, \dots$ is defined recursively as follows:

$$a_k = ra_{k-1} \text{ for all integers } k \geq 1$$

$$a_0 = a.$$

Use iteration to guess an explicit formula for this sequence.

$$a_0 = a$$

$$a_1 = ra_0$$

$$a_2 = ra_1 = r(ra_0) = r^2a_0$$

$$a_3 = ra_2 = r(r^2a_0) = r^3a_0$$

$$\vdots$$

$$a_n =$$

Definition 3.10

A sequence $a_0, a_1, a_2, \dots$ is called a geometric sequence if and only if there is a constant r such that

$$a_k = r^k a_{k-1} \text{ for arbitrary integer } k \geq 1.$$

Or, equivalently, $a_n = a_0 r^n$ for arbitrary integer $n \geq 0$.